



Cheap Energy Might Not Be Enough: A Trade Model of AI Compute Services

Second Submission: RR_EUR_2026_669

Innocent Bushayija Sano

reproducibility@worldbank.org

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This review verifies the reproducibility of the exhibits included in the paper "*Cheap Energy Might Not Be Enough: A Trade Model of AI Compute Services*".

Contents in this review:

1. Main findings
2. List of exhibits and reproducibility status
3. Reproduction Environment

Main findings

- **The exhibits in the paper did not meet the requirements for computational reproducibility, with 5 exhibits failing to reproduce accurately.**
- The code was successfully executed on a new computer after:
 1. Updating the working directory in line 40 of the do-file "run.do", and running it.
- The output demonstrates consistent stability across multiple runs. Specifically, executing the code two times consecutively yielded identical results.
- The code takes approximately 9 minutes to run.
- We conducted our reproducibility analysis based on the paper shared by the authors by email on June 03rd paper.
- **Verification Process and Data Handling:**
 - The reproducibility package relies on 1 type of data: open data.
 - Open data is included in the public reproducibility package.
 - Reviewers started verification from the pre-processed outputs shared by the authors, and all pre-processing steps are documented in the README and package. Reviewers also used data provided directly by the authors to conduct the reproducibility verification.
 - *data_hash_report.csv* lists the SHA256 hashes of all files in the Data folder to support data integrity checks. Users who acquire the forthcoming/accessible/limited-access/restricted data can use this file to verify that the data has not been altered.
- **Reproducibility Summary:**

- **Data:** All data sources are publicly available but not all are included in the reproducibility package.
- **Code:** All code files (from cleaning to analysis) are included in the reproducibility package.
- **Outputs:** All outputs are generated by code included in the reproducibility package.
- **Dependencies environment:** The reviewers created a new environment using the latest versions of dependencies available at the moment of the review.

List of exhibits and reproducibility status

Results in the Main Section of the Paper

- **Figure 1** Does not apply. Exhibit is a non-analytical output.
- **Table 1** Does not apply. Exhibit is a non-analytical output.
- **Table 2** Reproduced.
- **Table 3** Does not reproduce. See details below

Code						Paper					
iso3	country	c_j_raw	rank_raw	c_j_symme	rank_sy	Table 3. Country rankings under alternative cost and sovereignty specifications					
						(1) Raw Electricity			(2) Cost-Recovery		
						Country	P _i	Type	Country	P _j	I
IRN	Iran	1.40794	1	1.4820299		Iran	\$1.56	DD*	Kyrgyzstan	\$1.58	I
TKM	Turkmenis	1.41662	2	1.4728129		Turkmenistan	\$1.57	DD*	Ethiopia	\$1.59	I
ETH	Ethiopia	1.42737	3	1.4449130		Ethiopia	\$1.58	EE†	Kosovo	\$1.60	I
KGZ	Kyrgyzstan	1.42847	4	1.4284726		Kyrgyzstan	\$1.58	EE†	Canada	\$1.60	I
EGY	Egypt	1.43409	5	1.4731879		Egypt	\$1.58	EE†	Tajikistan	\$1.60	I
DZA	Algeria	1.43535	6	1.4665410		Algeria	\$1.59	EE†	Montenegro	\$1.60	I
CAN	Canada	1.44207	7	1.4478169		Canada	\$1.59	EE	China	\$1.61	I
RUS	Russia	1.44426	8	1.4636501		Russia	\$1.59	DD*	Ukraine	\$1.61	I
XKX	Kosovo	1.44686	10	1.4468578		South Africa	\$1.60	II	Argentina	\$1.61	I
TJK	Tajikistan	1.4544	12	1.4543722		Kosovo	\$1.60	IE†	Colombia	\$1.61	I
MNE	Montenegr	1.45474	13	1.4547660		Nigeria	\$1.60	II	Azerbaijan	\$1.61	I
CHN	China	1.45527	14	1.4552593		Tajikistan	\$1.60	II	Russia	\$1.61	I
UKR	Ukraine	1.45795	16	1.4579653		Montenegro	\$1.60	IE†	Norway	\$1.62	I
ARG	Argentina	1.46177	17	1.461788		China	\$1.61	II	Algeria	\$1.62	I
COL	Colombia	1.46289	18	1.4629034		Qatar	\$1.61	II	Vietnam	\$1.62	I
AZE	Azerbaijan	1.46308	19	1.4630638		Ukraine	\$1.61	IE†	Finland	\$1.62	I
NOR	Norway	1.46503	20	1.4660989		Argentina	\$1.61	II	Brazil	\$1.62	I
KAZ	Kazakhstan	1.46557	21	1.4818905		Colombia	\$1.61	II	Turkmenistan	\$1.62	I
						Azerbaijan	\$1.61	IE	Egypt	\$1.62	I
						Norway	\$1.62	IE	India	\$1.63	I
						Kazakhstan	\$1.62	II	Indonesia	\$1.63	I

Results in the Annex

Table A.1 Does not reproduce. See details below

Code									
iso3	country	p _E	theta	p _{UE}	c _{construct}	k _{bar}	g _{pu}	omega _{pci}	c _j total
IRN	Iran	0.005	31.2	1.324	0.04525	47430535	0.008081	1.40794	8.08E-05
TKM	Turkmenis	0.01	32.2	1.339	0.04919	2976545	0.00404	1.41662	4.04E-05
ETH	Ethiopia	0.03	26.5	1.252	0.04301	1699389	0.008081	1.42737	8.08E-05
KGZ	Kyrgyzstan	0.038	13.3	1.08	0.04168	1999500	0.00404	1.42847	4.04E-05
EGY	Egypt	0.038	31.7	1.331	0.04062	24427143	0.00404	1.43409	0.001212
DZA	Algeria	0.033	35.8	1.391	0.04514	12055143	0.020202	1.43535	0.000202
CAN	Canada	0.045	14.6	1.08	0.04999	82800428	2.5859	1.44207	0.025859
RUS	Russia	0.04	16.9	1.109	0.05515	1.48E+08	1.6161877	1.44426	0.016162
ZAF	South Africa	0.04	23.5	1.208	0.0543	29322857	0.202023	1.44619	0.00202
XXK	Kosovo	0.0468	21	1.17	0.05047	721143.43	0.00404	1.44686	4.04E-05
NGA	Nigeria	0.042	31.4	1.326	0.0559	4414526	0.080809	1.45294	0.000808
TJK	Tajikistan	0.074	16.8	1.108	0.03897	2190332	0.00404	1.4544	4.04E-05
MNE	Montenegro	0.0508	21.7	1.18	0.05473	406571.4	0.00404	1.45474	4.04E-05
CHN	China	0.079	20.9	1.169	0.03258	1.31E+09	25.778194	1.45527	25778194
QAT	Qatar	0.036	36.1	1.396	0.06314	7353826	0.064648	1.45638	0.000646
UKR	Ukraine	0.0568	22.5	1.192	0.05249	13564000	1.4141642	1.45795	0.001414
ARG	Argentina	0.06	22.3	1.189	0.05377	18357000	1.2121408	1.46177	0.001212
COL	Colombia	0.075	25.5	1.237	0.03987	11606571	1.2121408	1.46289	0.001212
AZE	Azerbaijan	0.062	24.4	1.221	0.05201	3433429	0.008081	1.46308	8.08E-05
NOR	Norway	0.0541	14.5	1.08	0.06607	18546000	32323754	1.46503	0.003232
KAZ	Kazakhstan	0.066	24.9	1.229	0.05073	14944286	0.080809	1.46557	0.000808
UZB	Uzbekistan	0.071	29.9	1.303	0.04456	10665399	0.020202	1.46737	0.000202
FIN	Finland	0.0571	16.4	1.101	0.06543	11480000	28283285	1.46748	0.002828
VNM	Vietnam	0.073	28	1.275	0.04663	37623483	0.202023	1.46984	0.00202

Paper

Table A1. Country-specific calibration parameters

Country	p ^E (\$/kWh)	q _i (°C)	PUE	Const _E (\$/W)	q _E (MW)	q _i (%)	q _E (\$/hr)	Cost Rec. p ^E (\$/kWh)
Kyrgyzstan	\$0.038	13.3	1.08	\$7.83	5	0.0	\$1.43	\$0.038
Ethiopia	\$0.030	26.5	1.25	\$8.08	10	0.0	\$1.43	\$0.069
Kosovo	\$0.047	21.0	1.17	\$9.48	5	0.0	\$1.43	\$0.047
Canada	\$0.043	14.6	1.08	\$9.39	3,200	2.6	\$1.44	\$0.083
Tajikistan	\$0.074	16.8	1.11	\$7.32	5	0.0	\$1.45	\$0.074
Montenegro	\$0.051	21.7	1.18	\$10.28	5	0.0	\$1.45	\$0.051
China	\$0.079	20.9	1.17	\$6.12	11,900	25.8	\$1.46	\$0.079
Ukraine	\$0.057	22.5	1.19	\$9.86	175	0.1	\$1.46	\$0.057
Argentina	\$0.060	22.3	1.19	\$10.10	150	0.1	\$1.46	\$0.060
Colombia	\$0.075	25.5	1.24	\$7.49	150	0.1	\$1.46	\$0.075
Azerbaijan	\$0.062	24.4	1.22	\$9.77	10	0.0	\$1.46	\$0.062
Russia	\$0.040	16.9	1.11	\$10.36	2,000	1.6	\$1.44	\$0.066
Norway	\$0.054	14.5	1.08	\$12.41	400	0.3	\$1.47	\$0.096
Algeria	\$0.033	35.8	1.39	\$8.48	25	0.0	\$1.44	\$0.066
Vietnam	\$0.073	28.0	1.27	\$8.76	350	0.2	\$1.47	\$0.073
Finland	\$0.057	16.4	1.10	\$12.29	350	0.3	\$1.47	\$0.062
Brazil	\$0.063	27.3	1.26	\$10.75	1,200	1.0	\$1.47	\$0.063
Turkmenistan	\$0.010	32.2	1.34	\$9.24	5	0.0	\$1.42	\$0.079
Egypt	\$0.038	31.7	1.33	\$7.63	150	0.1	\$1.43	\$0.089
India	\$0.090	30.2	1.31	\$6.64	3,600	2.9	\$1.48	\$0.090
Indonesia	\$0.067	26.4	1.25	\$11.21	500	0.4	\$1.48	\$0.067
Sweden	\$0.068	15.2	1.08	\$12.29	300	0.4	\$1.48	\$0.071
Kazakhstan	\$0.066	24.9	1.23	\$9.53	100	0.1	\$1.47	\$0.066
Iran	\$0.005	31.2	1.32	\$8.50	10	0.0	\$1.41	\$0.088
Georgia	\$0.091	19.4	1.15	\$9.58	15	0.0	\$1.48	\$0.091

Table A.2 Does not reproduce. See details below

Paper

Table A2. Country rankings under alternative cost and sovereignty specifications (all countries)

iso3	country	c_j_raw	rank_raw	c_j_symm	rank_symm	c_j_wacc	rank_wacc	income
KGZ	Kyrgyzstan	1.42847	4	1.4284726	1	1.8548099	68	LMIC
ETH	Ethiopia	1.42737	3	1.4449130	2	1.9606660	84	LIC
XXK	Kosovo	1.44686	10	1.4468578	3	1.7850787	34	UMIC
CAN	Canada	1.44207	7	1.4478169	4	1.6706754	1	HIC
TJK	Tajikistan	1.4544	12	1.4543722	5	1.8807095	72	LMIC
MNE	Montenegrr	1.45474	13	1.4547660	6	1.7929869	36	UMIC
CHN	China	1.45527	14	1.4552593	7	1.79348	37	UMIC
UKR	Ukraine	1.45795	16	1.4579653	8	1.796186	40	UMIC
ARG	Argentina	1.46177	17	1.461788	9	1.8000091	41	UMIC
COL	Colombia	1.46289	18	1.4629034	10	1.8011243	43	UMIC
AZE	Azerbaijan	1.46308	19	1.4630638	11	1.8012847	45	UMIC
RUS	Russia	1.44426	8	1.4636501	12	1.8018710	46	UMIC
NOR	Norway	1.46503	20	1.4660989	13	1.6889575	2	HIC
DZA	Algeria	1.43535	6	1.4665410	14	1.8047619	47	UMIC
VNM	Vietnam	1.46984	24	1.4698481	15	1.8961853	73	LMIC
FIN	Finland	1.46748	23	1.4710486	16	1.6939072	3	HIC
BRA	Brazil	1.47108	26	1.4710540	17	1.8092749	48	UMIC
TKM	Turkmenis	1.41662	2	1.4728129	18	1.8110338	49	UMIC
EGY	Egypt	1.43409	5	1.4731879	19	1.8995252	74	LMIC
IND	India	1.47581	29	1.4758136	20	1.9021508	75	LMIC
IDN	Indonesia	1.47638	30	1.47641	21	1.8146312	50	UMIC
SWE	Sweden	1.47517	28	1.4773415	22	1.7002000	4	HIC
KAZ	Kazakhstan	1.46557	21	1.4818905	23	1.820111	53	UMIC

(I) Raw Electricity				
Country	q _i	Rank	Type	q _i
Kyrgyzstan	\$1.58	4	EE	\$1.58
Ethiopia	\$1.58	3	EE	\$1.59
Kosovo	\$1.60	10	IE	\$1.60
Canada	\$1.59	7	EE	\$1.60
Tajikistan	\$1.60	12	II	\$1.60
Montenegro	\$1.60	13	IE	\$1.60
China	\$1.61	14	II	\$1.61
Ukraine	\$1.61	16	IE	\$1.61
Argentina	\$1.61	17	II	\$1.61
Colombia	\$1.61	18	II	\$1.61
Azerbaijan	\$1.61	19	IE	\$1.61
Russia	\$1.59	8	DD	\$1.61
Norway	\$1.62	20	IE	\$1.62
Algeria	\$1.59	6	EE	\$1.62
Vietnam	\$1.62	24	II	\$1.62
Finland	\$1.62	23	IE	\$1.62
Brazil	\$1.62	26	II	\$1.62
Turkmenistan	\$1.57	2	DD	\$1.62
Egypt	\$1.58	5	EE	\$1.62
India	\$1.63	29	II	\$1.63
Indonesia	\$1.63	30	IE	\$1.63
Sweden	\$1.63	28	II	\$1.63
Kazakhstan	\$1.62	21	II	\$1.63
Iran	\$1.56	1	DD	\$1.63
Georgia	\$1.63	31	II	\$1.63
Uzbekistan	\$1.62	22	II	\$1.63
Kenya	\$1.64	33	II	\$1.64
Turkey	\$1.64	34	II	\$1.64
Bosnia and Herz.	\$1.64	35	II	\$1.64

Table A.3 Does not reproduce. See details below

Code					Paper				
scena label	p_T	n_exporter hhi_T	rank_corr	top1	Table A3. Sensitivity of cost-recovery rankings to parameter variation				
					Scenario	Parameter change	Dev top 15	Max spread	Spearman ρ vs baseline
1 Baseline calibration	1.5920694	2	0.933919	1 KGZ	Baseline ($\rho=\$1.36$)	$\rho=\$1.36$	11	11.9%	1.00
2 Electricity price +\$0.01/kWh	1.5996294	2	0.933919	.99949188 KGZ					
3 Electricity price -\$0.01/kWh	1.5845094	2	0.933919	.99935509 KGZ					
4 GPU hardware cost +20% (\$30,000)	1.8636816	2	0.933919	1 KGZ	Low hardware share ($\rho=\$1.30$)	$\rho=\$1.30$ (-4.4%)	11	12.3%	1.00
5 GPU hardware cost -20% (\$20,000)	1.3204573	2	0.933919	1 KGZ	High hardware share ($\rho=\$1.42$)	$\rho=\$1.42$ (+4.4%)	11	11.4%	1.00
6 Cooling efficiency cap (PUE<=1.20)	1.5920694	2	0.933919	.99208520 KGZ					

Table A.4 Reproduced.

Table A.5 Reproduced.

Table A.6 Does not reproduce. See details below

Code			Paper		
scenario	npv	irr	Table A6. Sensitivity of investment returns to parameter variation		
			Scenario	NPV (\$M)	IRR
Base case	114795979.7395999	.1420939596044256	Base case	\$353	17.6%
GPU price -20%	732987622.4	0.242600308	GPU price -20%	\$930	27.6%
GPU price +20%	-503395663	0.062365462	GPU price +20%	\$-225	9.7%
Electricity +50%	67569664.58109258	.1354880704846904	Electricity +50%	\$305	16.9%
Electricity -25%	138409137.3	.1453900736220896	Electricity -25%	\$377	17.9%
Revenue price +5%	268918944.6443766	.1634910242492771	Revenue +5%	\$509	19.8%
Revenue price -5%	-39326985.17	.1204495169001152	Revenue -5%	\$198	15.4%
Utilization 80%	709711613.5461845	.2266471962238283	Utilization 80%	\$957	26.4%
Utilization 60%	-480119654.1	0.058347012	Utilization 60%	\$-253	9.0%
WACC 10%	346107906.2985052	.1420939596044256	WACC 10%	\$619	17.6%
WACC 16%	-107658062.8	.1420939596044256	WACC 16%	\$95	17.6%

Table A.7 Reproduced.

Table A.8 Does not apply. Exhibit is a non-analytical output.

Reproduction Environment

Paper exhibits were reproduced on a computer with the following specifications:

- OS: Windows 11 Enterprise
- Processor: Intel(R) Xeon(R) Gold 6226R CPU @ 2.90GHz (2.90 GHz) (2 processors)
- Memory available: 16.0 GB
- Software version: Stata 19.0 MP